

UNIT-I

Introduction to Machine Learning: Brief Introduction, Learning Paradigms, Statistical Decision Theory – Regression and Classification, Bias-Variance.

Course Objectives: The students will learn the following

1. Understand the fundamentals of Machine Learning and its various learning paradigms.
2. Comprehend the Statistical Decision Theory and its application in Regression and Classification models.
3. Analyze the Bias-Variance trade-off and its significance in Machine Learning models.
4. Build and interpret Regression and Classification models such as Linear Regression, Logistic Regression, and Naïve-Bayes.
5. Understand and implement different clustering algorithms such as Partitional Clustering, Hierarchical Clustering, and Density-based Clustering.

Course Outcomes: Upon the successful completion of the course, the students will able to:

- CO1. Apply the basic principles and concepts of Machine Learning and its various learning paradigms.
- CO2. Apply Statistical Decision Theory in Regression and Classification models to make informed decisions.
- CO3. Implement Bias-Variance trade-off and its importance in Machine Learning models.
- CO4. Interpret Regression and Classification models such as Linear Regression, Logistic Regression, and Naïve-Bayes.
- CO5. Implement clustering algorithms such as Partitional Clustering, Hierarchical Clustering, and Density-based Clustering, and develop an understanding of their applicability in real-world scenarios.

Learning Resources

Textbooks	Unit I	Unit II	Unit III	Unit IV	Unit V
1. Introduction to Machine Learning, Ethem Alpaydin 2nd Edition, MIT Press 2000.	C	C	C	C	C
2. Machine Learning, Tom M. Mitchell, McGraw hill, 1997, ISBN: 0-07-042807-7.			C	C	C

1 Unit - I

1.1 Brief Introduction

Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, coined the term “Machine Learning ” in 1959 while at IBM. He defined machine learning as “the field of study that gives computers the ability to learn without being explicitly programmed”.

Machine learning is a subfield of artificial intelligence that involves the development of algorithms and statistical models that enable computers to improve their performance in tasks through experience. These algorithms and models are designed to learn from data and make predictions or decisions without explicit instructions. There are several types of machine learning, including supervised learning, unsupervised learning, and reinforcement learning. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data. Reinforcement learning involves training a model through trial and error.

Definition of learning: A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks T, as measured by P, improves with experience E.

Examples

- Handwriting recognition learning problem
 - Task T : Recognizing and classifying handwritten words within images
 - Performance P : Percent of words correctly classified
 - Training experience E : A dataset of handwritten words with given classifications
- A robot driving learning problem
 - Task T : Driving on highways using vision sensors
 - Performance P : Average distance traveled before an error
 - Training experience E : A sequence of images and steering commands recorded while observing a human driver

This notes is collected from sources [credits](#) Machine learning applications are used in finance, health-care, hospitality, government, and beyond, is already in regular use. Businesses are beginning to see the benefits of using machine learning tools to improve their processes, gain valuable insights from unstructured data, and automate tasks that would otherwise require hours of tedious, manual work (which usually produces much less accurate results).

For example, UberEats uses machine learning to estimate optimum times for drivers to pick up food orders, while Spotify leverages machine learning to offer personalized content and personalized marketing. And Dell uses machine learning text analysis to save hundreds of hours analyzing thousands of employee surveys to listen to the voice of employee (VoE) and improve employee satisfaction.

The Google Maps predicts peaks in traffic and Netflix creates personalized movie recommendations, even informs the creation of new content

A few common machine learning use cases and applications as follows:

- Social Media Monitoring
- Customer Service & Customer Satisfaction
- Image Recognition
- Virtual Assistants
- Product Recommendations
- Stock Market Trading
- Medical Diagnosis

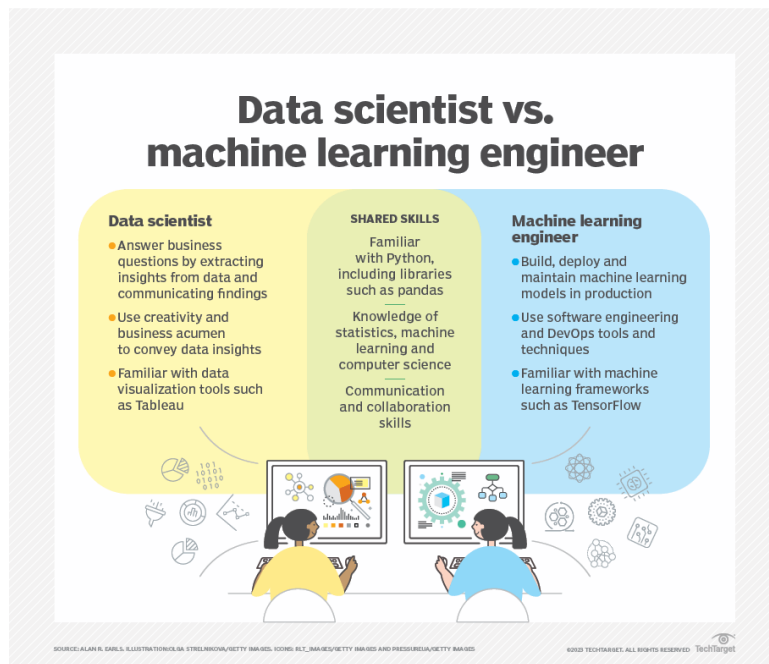


Figure 1: Although the two disciplines are interconnected, data science and machine learning have some important differences.

Image credits

Social Media Monitoring Using machine learning one can monitor mentions of your brand on social media and immediately identify if customers require urgent attention. By detecting mentions from angry customers, in real-time, one can automatically tag customer feedback and respond right away. You might also want to analyze customer support interactions on social media and gauge customer satisfaction (CSAT), to see how well your team is performing.

Natural Language Processing gives machines the ability to break down spoken or written language much like a human would, to process “natural” language, so machine learning can handle text from practically any source.

Customer Service & Customer Satisfaction Machine learning allows one to integrate powerful text analysis tools with customer support tools, so one can analyze your emails, live chats, and all manner of internal data on the go. You can use machine learning to tag support tickets and route them to the correct teams or auto-respond to common queries so one never leave a customer in the cold. Furthermore, using machine learning to set up a voice of customer (VoC) program and a customer feedback loop will ensure that one follow the customer journey from start to finish to improve the customer experience (CX), decrease customer churn, and, ultimately, increase your profits.

Image Recognition Image recognition is helping companies identify and classify images. For example, facial recognition technology is being used as a form of identification, from unlocking phones to making payments.

Self-driving cars also use image recognition to perceive space and obstacles. For example, they can learn to recognize stop signs, identify intersections, and make decisions based on what they see.

Virtual Assistants Virtual assistants, like Siri, Alexa, Google Now, all make use of machine learning to automatically process and answer voice requests. They quickly scan information, remember related queries, learn from previous interactions, and send commands to other apps, so they can collect information and deliver the most effective answer.

Customer support teams are already using virtual assistants to handle phone calls, automatically route support tickets, to the correct teams, and speed up interactions with customers via computer-generated responses.

Product Recommendations Association rule-learning is a machine learning technique that can be used to analyze purchasing habits at the supermarket or on e-commerce sites. It works by searching for relationships between variables and finding common associations in transactions (products that consumers usually buy together). This data is then used for product placement strategies and similar product recommendations.

Associated rules can also be useful to plan a marketing campaign or analyze web usage.

Stock Market Trading Machine learning algorithms can be trained to identify trading opportunities, by recognizing patterns and behaviors in historical data. Humans are often driven by emotions when it comes to making investments, so sentiment analysis with machine learning can play a huge role in identifying good and bad investing opportunities, with no human bias, whatsoever. They can even save time and allow traders more time away from their screens by automating tasks.

Medical Diagnosis The ability of machines to find patterns in complex data is shaping the present and future. Take machine learning initiatives during the COVID-19 outbreak, for instance. AI tools have helped predict how the virus will spread over time, and shaped how we control it. It's also helped diagnose patients by analyzing lung CTs and detecting fevers using facial recognition, and identified patients at a higher risk of developing serious respiratory disease.

1.2 Learning paradigms

Learning paradigms tell different approaches of machine learning methods(algorithms).

1.2.1 classification based on nature of learning signal

The machine learning implementations are classified into four major categories, depending on the nature of the learning “signal” or “response” available to a learning system which are as follows:

Supervised learning: Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs. The given data is labeled. Both classification and regression problems are supervised learning problems. Supervised learning algorithms are used for several tasks, including the following:

- **Binary classification.** Divides data into two categories.
- **Multiclass classification.** Chooses between more than two types of answers.
- **Ensembling.** Combines the predictions of multiple ML models to produce a more accurate prediction.
- **Regression modeling.** Predicts continuous values based on relationships within data.

Unsupervised learning: Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses. In unsupervised learning algorithms, classification or categorization is not included in the observations. Unsupervised learning algorithms are good for the following tasks:

- **Clustering.** Splitting the data set into groups based on similarity using clustering algorithms.
- **Anomaly detection.** Identifying unusual data points in a data set using anomaly detection algorithms.
- **Association rule mining.** Discovering sets of items in a data set that frequently occur together using association rule mining.

- **Dimensionality reduction.** Decreasing the number of variables in a data set using dimensionality reduction techniques.

Reinforcement learning Reinforcement learning is the problem of getting an agent to act in the world so as to maximize its rewards.

A learner is not told what actions to take as in most forms of machine learning but instead must discover which actions yield the most reward by trying them. For example — Consider teaching a dog a new trick: we cannot tell him what to do, what not to do, but we can reward/punish it if it does the right/wrong thing. Reinforcement learning is often used in the following areas:

- **Robotics.** Robots learn to perform tasks in the physical world.
- **Video gameplay.** Teaches bots to play video games.
- **Resource management.** Helps enterprises plan allocation of resources.

Semi-supervised learning Where an incomplete training signal is given: a training set with some (often many) of the target outputs missing. There is a special case of this principle known as Transduction where the entire set of problem instances is known at learning time, except that part of the targets are missing. Semi-supervised learning is an approach to machine learning that combines small labeled data with a large amount of unlabeled data during training. Semi-supervised learning falls between unsupervised learning and supervised learning.

1.2.2 classification based on required output

Another categorization of machine-learning tasks arises when one considers the desired output of a machine-learned system:

Classification: When inputs are divided into two or more classes, the learner must produce a model that assigns unseen inputs to one or more (multi-label classification) of these classes. This is typically tackled in a supervised way. Spam filtering is an example of classification, where the inputs are email (or other) messages and the classes are “spam” and “not spam”.

Regression: Which is also a supervised problem, A case when the outputs are continuous rather than discrete.

Clustering: When a set of inputs is to be divided into groups. Unlike in classification, the groups are not known beforehand, making this typically an unsupervised task.

Machine Learning comes into the picture when problems cannot be solved using typical approaches. ML algorithms combined with new computing technologies promote scalability and improve efficiency. Modern ML models can be used to make predictions ranging from outbreaks of disease to the rise and fall of stocks.

1.3 Statistical decision theory

1.4 Regression and classification

1.4.1 Regression

Regression Analysis is a statistical process for estimating the relationships between the dependent variables or criterion variables and one or more independent variables or predictors. A regression problem is when the output variable is a real or continuous value, such as “salary” or “weight”. the development of the machine learning domain regression analysis techniques have gained popularity as

well as developed manifold from just $y = mx + c$. There are several types of regression techniques,

1. Linear Regression
2. Polynomial Regression
3. Stepwise Regression
4. Decision Tree Regression
5. Random Forest Regression
6. Support Vector Regression
7. Ridge Regression
8. Lasso Regression
9. ElasticNet Regression
10. Bayesian Linear Regression

Linear regression is used for predictive analysis. Linear regression is a linear approach for modeling the relationship between the criterion or the scalar response and the multiple predictors or explanatory variables. Linear regression focuses on the conditional probability distribution of the response given the values of the predictors. For linear regression, there is a danger of overfitting. The formula for linear regression is:

$$y = \beta_0 + \beta_1 x_1 + \epsilon$$

where,

- β_0 is the model weights or parameters or intercept
- β_1 is the bias or slope
- x_1 is the independent or predictor or feature or covariate or explanatory variable
- y is the dependent or target or response variable.
- ϵ error term

Here, a linear regression model is instantiated to fit a linear relationship between input features (X) and target values (y).

```
from sklearn.linear_model import LinearRegression
```

```
# Create a linear regression model
model = LinearRegression()
```

```
# Fit the model to the data
model.fit(X, y)
```

```
# Predict the response for a new data point
y_pred = model.predict(X_new)
```

1.4.2 Classification

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc. Classes can be called as targets/labels or categories. Classification algorithm is a Supervised learning technique, hence it takes labeled input data, which means it contains input with the corresponding output. In classification algorithm, a discrete output function(y) is mapped to input variable(x).

The algorithm which implements the classification on a dataset is known as a classifier. There are two types of Classifications:

Binary Classifier: If the classification problem has only two possible outcomes, then it is called as Binary Classifier. Examples: YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.

Multi-class Classifier: If a classification problem has more than two outcomes, then it is called as Multi-class Classifier. Example: Classifications of types of crops, Classification of types of music.

In the classification problems, there are two types of learners:

Lazy Learners: Lazy Learner firstly stores the training dataset and wait until it receives the test dataset. In Lazy learner case, classification is done on the basis of the most related data stored in the training dataset. It takes less time in training but more time for predictions. Example: K-NN algorithm, Case-based reasoning

Eager Learners: Eager Learners develop a classification model based on a training dataset before receiving a test dataset. Opposite to Lazy learners, Eager Learner takes more time in learning, and less time in prediction. Example: Decision Trees, Naïve Bayes, ANN.

Classification Algorithms can be further divided into the Mainly two category:

- Linear Models
 - Logistic Regression
 - Support Vector Machines
- Non-linear Models
 - K-Nearest Neighbours
 - Kernel SVM
 - Naïve Bayes
 - Decision Tree Classification
 - Random Forest Classification

1.5 Bias-Variance

Machine learning is a branch of Artificial Intelligence, which allows machines to perform data analysis and make predictions. However, if the machine learning model is not accurate, it can make predictions errors, and these prediction errors are usually known as Bias and Variance. In machine learning, these errors will always be present as there is always a slight difference between the model predictions and actual predictions. The main aim of ML/data science analysts is to reduce these errors in order to get more accurate results. In machine learning, an error is a measure of how accurately an algorithm can make predictions for the previously unknown dataset. On the basis of these errors, the machine learning model is selected that can perform best on the particular dataset. There are mainly two types of errors in machine learning, which are:

Reducible errors: These errors can be reduced to improve the model accuracy. Such errors can further be classified into bias and Variance. Bias and Variance in Machine Learning

Irreducible errors: These errors will always be present in the model regardless of which algorithm has been used. The cause of these errors is unknown variables whose value can't be reduced.

1.5.1 Bias

In general, a machine learning model analyses the data, find patterns in it and make predictions. While training, the model learns these patterns in the dataset and applies them to test data for prediction. While making predictions, a difference occurs between prediction values made by the model and actual values/expected values, and this difference is known as bias errors or Errors due to bias. It can be defined as an inability of machine learning algorithms such as Linear Regression to capture the true relationship between the data points. Each algorithm begins with some amount of bias because bias occurs from assumptions in the model, which makes the target function simple to learn. A model has either:

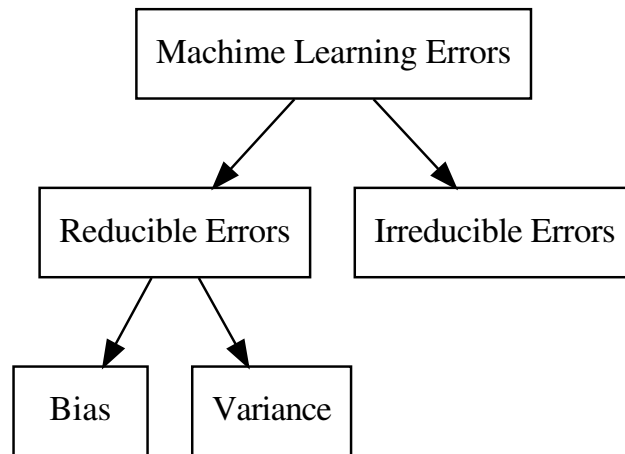


Figure 2: Classification of ML errors

Low Bias: A low bias model will make fewer assumptions about the form of the target function.

High Bias: A model with a high bias makes more assumptions, and the model becomes unable to capture the important features of our dataset. A high bias model also cannot perform well on new data.

Generally, a linear algorithm has a high bias, as it makes them learn fast. The simpler the algorithm, the higher the bias it has likely to be introduced. Whereas a nonlinear algorithm often has low bias.

Some examples of machine learning algorithms with low bias are Decision Trees, k-Nearest Neighbours and Support Vector Machines. At the same time, an algorithm with high bias is Linear Regression, Linear Discriminant Analysis and Logistic Regression.

Ways to reduce High Bias: High bias mainly occurs due to a much simple model. Below are some ways to reduce the high bias:

- Increase the input features as the model is underfitted.
- Decrease the regularization term.
- Use more complex models, such as including some polynomial features.

1.5.2 Variance

The variance would specify the amount of variation in the prediction if the different training data was used. In simple words, variance tells that how much a random variable is different from its expected value. Ideally, a model should not vary too much from one training dataset to another, which means the algorithm should be good in understanding the hidden mapping between inputs and output variables. Variance errors are either of low variance or high variance.

Low variance means there is a small variation in the prediction of the target function with changes in the training data set. At the same time, High variance shows a large variation in the prediction of the target function with changes in the training dataset.

A model that shows high variance learns a lot and perform well with the training dataset, and does not generalize well with the unseen dataset. As a result, such a model gives good results with the training dataset but shows high error rates on the test dataset.

Since, with high variance, the model learns too much from the dataset, it leads to overfitting of the model. A model with high variance has the below problems:

A high variance model leads to overfitting. Increase model complexities.

Usually, nonlinear algorithms have a lot of flexibility to fit the model, have high variance. Some examples of machine learning algorithms with low variance are, Linear Regression, Logistic Regression,

and Linear discriminant analysis. At the same time, algorithms with high variance are decision tree, Support Vector Machine, and K-nearest neighbours. Ways to Reduce High Variance:

- Reduce the input features or number of parameters as a model is overfitted.
- Do not use a much complex model.
- Increase the training data.
- Increase the Regularization term.

Different Combinations of Bias-Variance

There are four possible combinations of bias and variances, which are represented by the below diagram: Bias and Variance in Machine Learning

Low-Bias, Low-Variance: The combination of low bias and low variance shows an ideal machine learning model. However, it is not possible practically.

Low-Bias, High-Variance: With low bias and high variance, model predictions are inconsistent and accurate on average. This case occurs when the model learns with a large number of parameters and hence leads to an overfitting

High-Bias, Low-Variance: With High bias and low variance, predictions are consistent but inaccurate on average. This case occurs when a model does not learn well with the training dataset or uses few numbers of the parameter. It leads to underfitting problems in the model.

High-Bias, High-Variance: With high bias and high variance, predictions are inconsistent and also inaccurate on average.

1.5.3 How to identify High variance or High Bias?

High variance can be identified if the model has: Low training error and high test error.

High Bias can be identified if the model has: High training error and the test error is almost similar to training error.

1.6 Bias-variance trade-off

While building the machine learning model, it is really important to take care of bias and variance in order to avoid overfitting and underfitting in the model. If the model is very simple with fewer parameters, it may have low variance and high bias. Whereas, if the model has a large number of parameters, it will have high variance and low bias. So, it is required to make a balance between bias and variance errors, and this balance between the bias error and variance error is known as the Bias-Variance trade-off. Bias and Variance in Machine Learning

For an accurate prediction of the model, algorithms need a low variance and low bias. But this is not possible because bias and variance are related to each other:

- If we decrease the variance, it will increase the bias.
- If we decrease the bias, it will increase the variance.

Bias-Variance trade-off is a central issue in supervised learning. Ideally, we need a model that accurately captures the regularities in training data and simultaneously generalizes well with the unseen dataset. Unfortunately, doing this is not possible simultaneously. Because a high variance algorithm may perform well with training data, but it may lead to overfitting to noisy data. Whereas, high bias algorithm generates a much simple model that may not even capture important regularities in the data. So, we need to find a sweet spot between bias and variance to make an optimal model.

Hence, the Bias-Variance trade-off is about finding the sweet spot to make a balance between bias and variance errors.

1.7 Underfitting and Overfitting

1.8 Underlying content required for better understanding

1.8.1 Challenges of machine learning

1.8.2 Performance of metrics

Metrics are different from loss functions. Loss functions show a measure of model performance. They're used to train a machine learning model (using some kind of optimization like Gradient Descent), and they're usually differentiable in the model's parameters.

Metrics are used to monitor and measure the performance of a model (during training and testing), and don't need to be differentiable. However, if, for some tasks, the performance metric is differentiable, it can also be used as a loss function (perhaps with some regularizations added to it), such as MSE.

Task	Measure
Regression	Error
Classification	Error
Clustering	Scatter / Purity
Associations	Support / Confidence
Reinforcement learning	cost / reward

Table 1: Measures used to evaluate ML models

1.8.3 Regression

In order to evaluate Regression models, we'll discuss these metrics in detail:

1. Mean Squared Error (MSE),
2. Mean Absolute Error (MAE),
3. Root Mean Squared Error (RMSE),
4. R^2 (R-Squared).

y_i : ground-truth value, $\hat{y}_i$: predicted value from a model, N : number of datums

Mean squared error Mean squared error finds the average of the squared difference between the target value and the value predicted by the regression model.

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Few key points related to MSE:

1. It's differentiable, so it can be optimized better.
2. It penalizes even small errors by squaring them, which essentially leads to an overestimation of how bad the model is.
3. Error interpretation has to be done with squaring factor(scale) in mind. For example in our Boston Housing regression problem, we got MSE=21.89 which primarily corresponds to (Prices)².
4. Due to the squaring factor, it's fundamentally more prone to outliers than other metrics.

Mean Absolute Error (MAE) Mean Absolute Error is the average of the difference between the ground truth and the predicted values. Mathematically, its represented as :

$$MAE = \frac{1}{N} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Few key points for MAE

1. It's more robust towards outliers than MSE, since it doesn't exaggerate errors.
2. Error interpretation needs no second thoughts, as it perfectly aligns with the original degree of the variable.
3. MAE is non-differentiable as opposed to MSE, which is differentiable.
4. It gives us a measure of how far the predictions were from the actual output. However, since MAE uses absolute value of the residual, it doesn't give us an idea of the direction of the error, i.e. whether we're under-predicting or over-predicting the data.

Root Mean Squared Error Root Mean Squared Error corresponds to the square root of the average of the squared difference between the target value and the value predicted by the regression model. Basically, $\sqrt{\text{MSE}}$. Mathematically it can be represented as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Few key points related to RMSE:

1. It retains the differentiable property of MSE.
2. It handles the penalization of smaller errors done by MSE by square rooting it.
3. Error interpretation can be done smoothly, since the scale is now the same as the random variable.
4. Since scale factors are essentially normalized, it's less prone to struggle in the case of outliers.

R² Coefficient of determination R² Coefficient of determination actually works as a post metric, meaning it's a metric that's calculated using other metrics.

The point of even calculating this coefficient is to answer the question "How much (what %) of the total variation in Y(target) is explained by the variation in X(regression line)"

This is calculated using the sum of squared errors. Let's go through the formulation to understand it better.

Total variation in Y (Variance of Y):

$$SE(\bar{Y}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Percentage of variation described the regression line:

$$\frac{SE(line)}{SE(\bar{Y})}$$

Subsequently, the percentage of variation described the regression line:

$$1 - \frac{SE(line)}{SE(\bar{Y})}$$

Finally, we have our formula for the coefficient of determination, which can tell us how good or bad the fit of the regression line is:

$$coef R^2 = 1 - \frac{SE(line)}{SE(\bar{Y})}$$

Few intuitions related to R² results:

1. If the sum of Squared Error of the regression line is small $\implies R^2$ will be close to 1 (Ideal), meaning the regression was able to capture 100% of the variance in the target variable.
2. Conversely, if the sum of squared error of the regression line is high $\implies R^2$ will be close to 0, meaning the regression wasn't able to capture any variance in the target variable.
3. You might think that the range of R^2 is (0,1) but it's actually $(-\infty, 1)$ because the ratio of squared errors of the regression line and mean can surpass the value 1 if the squared error of regression line is too high ($>$ squared error of the mean).

Adjusted R^2 The Vanilla R^2 method suffers from some demons, like misleading the researcher into believing that the model is improving when the score is increasing but in reality, the learning is not happening. This can happen when a model overfits the data, in that case the variance explained will be 100% but the learning hasn't happened. To rectify this, R^2 is adjusted with the number of independent variables.

Adjusted R^2 is always lower than R^2 , as it adjusts for the increasing predictors and only shows improvement if there is a real improvement.

$$R_a^2 = 1 - \left[\left(\frac{n-1}{n-k-1} \right) \cdot (1 - R^2) \right]$$

1.8.4 classification

Classification models have discrete output, so we need a metric that compares discrete classes in some form. Classification Metrics evaluate a model's performance and tell one how good or bad the classification is, but each of them evaluates it in a different way.

1. Confusion Matrix (not a metric but fundamental to others)
2. Accuracy
3. Precision and Recall
4. F1-score
5. AU-ROC

Confusion matrix A confusion matrix is a tabular representation of prediction outcomes of any classifier, which is used to describe the performance of the classification model on a set of test data when true values are known.

A typical confusion matrix for a binary classifier looks like the below image(However, it can be extended to use for classifiers with more than two classes).

Predicted values→ True values↓	Positive	Negative
True	TP	TN
False	FP	FN

Table 2: Confusion matrix for binary classification

In general, the table is divided into four terminologies, which are as follows:

1. **True Positive(TP):** In this case, the prediction outcome is true, and it is true in reality, also.
2. **True Negative(TN):** in this case, the prediction outcome is false, and it is false in reality, also.
3. **False Positive(FP):** In this case, prediction outcomes are true, but they are false in actuality.

4. **False Negative(FN):** In this case, predictions are false, and they are true in actuality.

Accuracy Classification accuracy is perhaps the simplest metric to use and implement and is defined as the number of correct predictions divided by the total number of predictions, multiplied by 100.

$$\begin{aligned} \text{Accuracy} &= \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \\ &= \frac{TP + FN}{TP + TN + FP + FN} \end{aligned} \quad (1)$$

This may be calculated in python using

```
from sklearn.metrics import accuracy_score
```

Precision is the ratio of true positives and total positives predicted:

$$\begin{aligned} \text{Precision} &= \frac{\text{Number true positives}}{\text{Total number of positives predicted}} \\ &= \frac{TP}{TP + FP} \end{aligned} \quad (2)$$

Precision values always in the range (0, 1) i.e. $0 < P < 1$

The precision metric focuses on Type-I errors(FP). A Type-I error occurs when we reject a true null Hypothesis(H_0). So, in this case, Type-I error is incorrectly labeling cancer patients as non-cancerous.

A precision score towards 1 will signify that your model didn't miss any true positives, and is able to classify well between correct and incorrect labeling of cancer patients. What it cannot measure is the existence of Type-II error, which is false negatives – cases when a non-cancerous patient is identified as cancerous.

A low precision score (< 0.5) means your classifier has a high number of false positives which can be an outcome of imbalanced class or untuned model hyperparameters. In an imbalanced class problem, one has to prepare your data beforehand with over/under-sampling or focal loss in order to curb FP/FN.

Recall/Sensitivity/Hit-Rate A Recall is essentially the ratio of true positives to all the positives in ground truth.

$$\begin{aligned} \text{Recall} &= \frac{\text{Number true positives}}{\text{The total positive predictions (True Positive and False Positive)}} \\ &= \frac{TP}{TP + FN} \end{aligned} \quad (3)$$

Recall values always in the range (0, 1) i.e. $0 < R < 1$

The recall metric focuses on type-II errors(FN). A type-II error occurs when we accept a false null hypothesis(H_0). So, in this case, type-II error is incorrectly labeling non-cancerous patients as cancerous.

Recall towards 1 will signify that your model didn't miss any true positives, and is able to classify well between correctly and incorrectly labeling of cancer patients.

What it cannot measure is the existence of type-I error which is false positives i.e the cases when a cancerous patient is identified as non-cancerous.

A low recall score (< 0.5) means your classifier has a high number of false negatives which can be an outcome of imbalanced class or untuned model hyperparameters. In an imbalanced class problem, one has to prepare your data beforehand with over/under-sampling or focal loss in order to curb FP/FN.

Precision-Recall tradeoff To improve performance of a model, you can either improve precision or recall – but not both! If you try to reduce cases of non-cancerous patients being labeled as cancerous (FN/type-II), no direct effect will take place on cancerous patients being labeled as non-cancerous.

F1-score The F1-score metric uses a combination of precision and recall. In fact, the F1 score is the harmonic mean of the two.

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

The formula of the two essentially is: a high F1 score symbolizes a high precision as well as high recall. It presents a good balance between precision and recall and gives good results on imbalanced classification problems. A low F1 score tells you (almost) nothing - it only tells you about performance at a threshold. Low recall means we didn't try to do well on very much of the entire test set. Low precision means that, among the cases we identified as positive cases, we didn't get many of them right.

But low F1 doesn't say which cases. High F1 means we likely have high precision and recall on a large portion of the decision (which is informative). With low F1, it's unclear what the problem is (low precision or low recall?), and whether the model suffers from type-I or type-II error.

F1 is widely used, and considered a fine metric to converge onto a decision, but not without some tweaks. Using FPR (false positive rates) along with F1 will help curb type-I errors, and you'll get an idea about the villain behind your low F1 score.

AUROC (Area under Receiver operating characteristics curve) Better known as AUC-ROC score/curves. It makes use of true positive rates(TPR) and false positive rates(FPR).

Intuitively TPR/recall corresponds to the proportion of positive data points that are correctly considered as positive, with respect to all positive data points. In other words, the higher the TPR, the fewer positive data points we will miss. Intuitively FPR/fallout corresponds to the proportion of negative data points that are mistakenly considered as positive, with respect to all negative data points. In other words, the higher the FPR, the more negative data points we will misclassify.

A no-skill classifier is one that can't discriminate between the classes, and would predict a random class or a constant class in all cases. The no-skill line changes based on the distribution of the positive to negative classes. It's a horizontal line with the value of the ratio of positive cases in the dataset. For a balanced dataset, it's 0.5.

The area equals the probability that a randomly chosen positive example ranks above (is deemed to have a higher probability of being positive than negative) a randomly chosen negative example.

So, high ROC simply means that the probability of a randomly chosen positive example is indeed positive. High ROC also means your algorithm does a good job at ranking test data, with most negative cases at one end of a scale and positive cases at the other.

ROC curves aren't a good choice when your problem has a huge class imbalance. The reason for this is not straightforward but can be intuitively seen using the formulas, you can read more about it here. You can still use them in that scenario after processing an imbalance set, or using focal loss techniques.

The AUROC metric has no use other than academic research, and comparing different classifiers.

MACHINE LEARNING

UNIT-1

Machine Learning:

Machine learning is a field of artificial intelligence that focuses on enabling computers to learn and improve from experience without being explicitly programmed. It involves developing algorithms and models that can learn from data to make predictions or decisions.

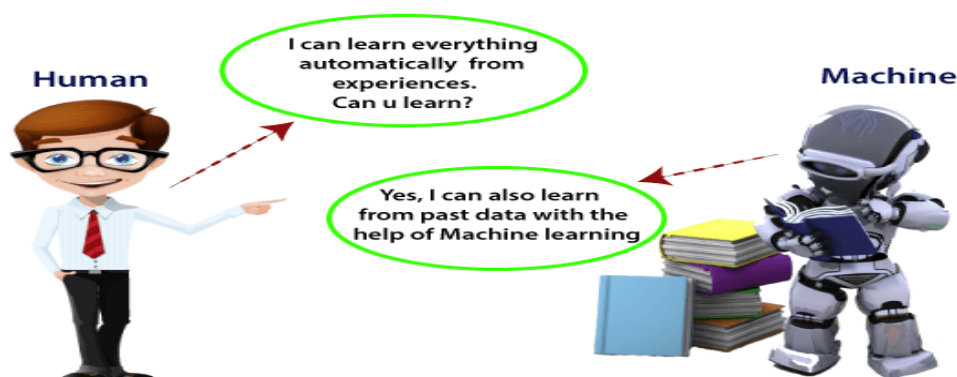


Machine Learning

A rapidly developing field of technology, machine learning allows computers to automatically learn from previous data. For building mathematical models and making predictions based on historical data or information, machine learning employs a variety of algorithms. It is currently being used for a variety of tasks, including speech recognition, email filtering, auto-tagging on Facebook, a recommender system, and image recognition.

What is Machine Learning

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.



Introduction to Machine Learning

1. Brief Introduction:

A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences. Arthur Samuel first used the term "machine learning" in 1959. It could be summarized as follows:

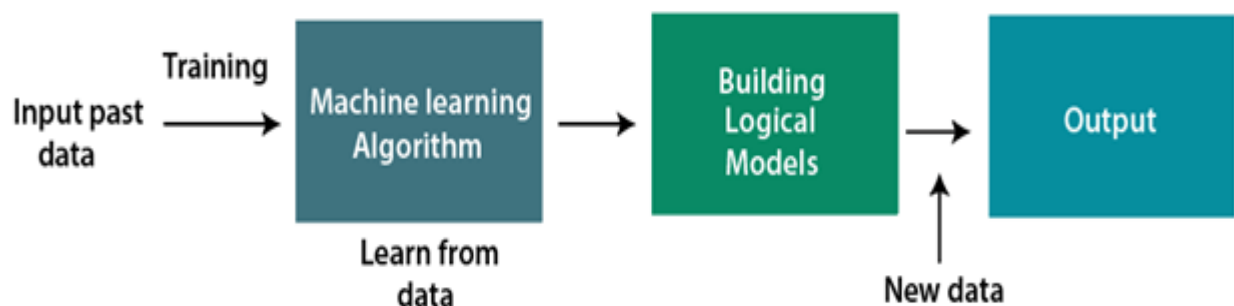
Without being explicitly programmed, machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things.

Machine learning algorithms create a mathematical model that, without being explicitly programmed, aids in making predictions or decisions with the assistance of sample historical data, or training data. For the purpose of developing predictive models, machine learning brings together statistics and computer science. Algorithms that learn from historical data are either constructed or utilized in machine learning. The performance will rise in proportion to the quantity of information we provide.

How does Machine Learning work

A machine learning system builds prediction models, learns from previous data, and predicts the output of new data whenever it receives it. The amount of data helps to build a better model that accurately predicts the output, which in turn affects the accuracy of the predicted output.

Let's say we have a complex problem in which we need to make predictions. Instead of writing code, we just need to feed the data to generic algorithms, which build the logic based on the data and predict the output. Our perspective on the issue has changed as a result of machine learning. The Machine Learning algorithm's operation is depicted in the following block diagram:



Features of Machine Learning:

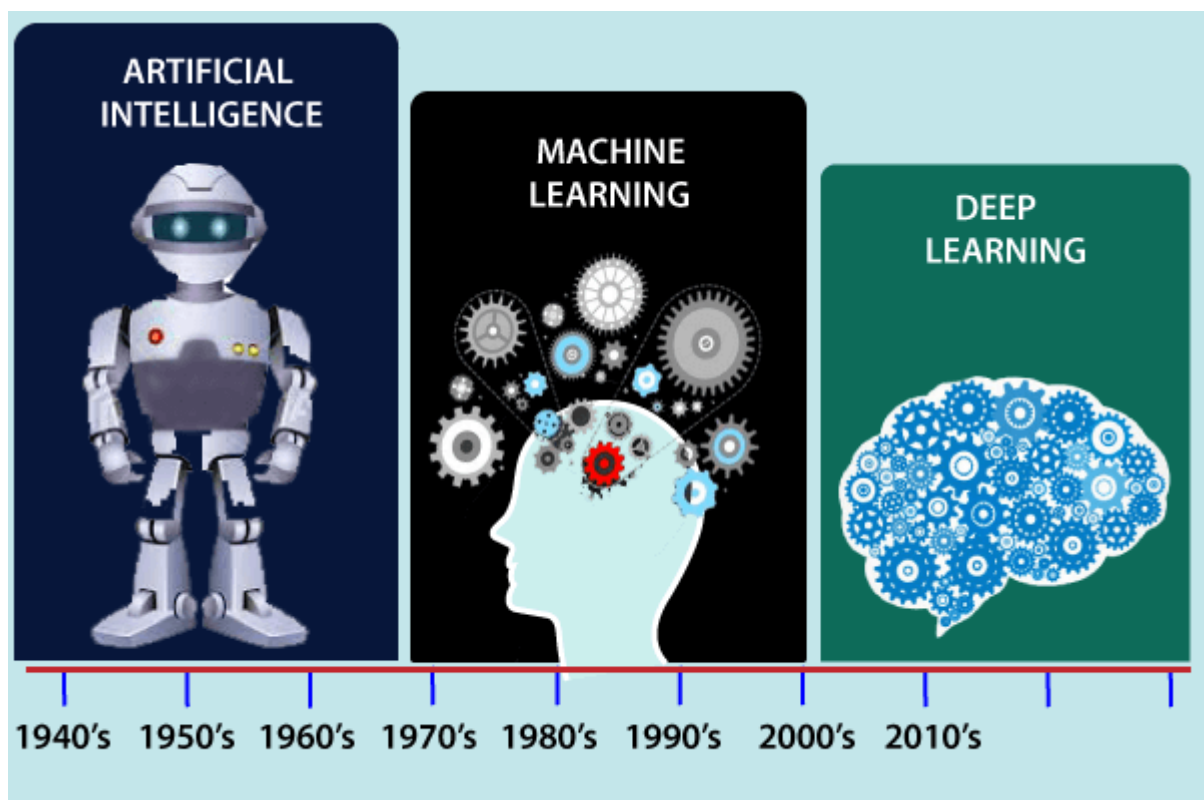
- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

key points which show the importance of Machine Learning:

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance
- Finding hidden patterns and extracting useful information from data.

History of Machine Learning

Before some years (about 40-50 years), machine learning was science fiction, but today it is the part of our daily life. Machine learning is making our day to day life easy from self-driving cars to Amazon virtual assistant "Alexa". However, the idea behind machine learning is so old and has a long history. Below some milestones are given which have occurred in the history of machine learning:



The era of stored program computers:

- **1940:** In 1940, the first manually operated computer, "ENIAC" was invented, which was the first electronic general-purpose computer. After that stored program computer such as EDSAC in 1949 and EDVAC in 1951 were invented.
- **1943:** In 1943, a human neural network was modeled with an electrical circuit. In 1950, the scientists started applying their idea to work and analyzed how human neurons might work.

Computer machinery and intelligence:

- **1950:** In 1950, Alan Turing published a seminal paper, "**Computer Machinery and Intelligence**," on the topic of artificial intelligence. **In his paper, he asked, "Can machines think?"**

Machine intelligence in Games:

- **1952:** Arthur Samuel, who was the pioneer of machine learning, created a program that helped an IBM computer to play a checkers game. It performed better more it played.
- **1959:** In 1959, the term "Machine Learning" was first coined by **Arthur Samuel**.

The first "AI" winter:

- The duration of 1974 to 1980 was the tough time for AI and ML researchers, and this duration was called as **AI winter**.
- In this duration, failure of machine translation occurred, and people had reduced their interest from AI, which led to reduced funding by the government to the researches.

Machine Learning from theory to reality

- **1959:** In 1959, the first neural network was applied to a real-world problem to remove echoes over phone lines using an adaptive filter.

- **1985:** In 1985, Terry Sejnowski and Charles Rosenberg invented a neural network **NETtalk**, which was able to teach itself how to correctly pronounce 20,000 words in one week.
- **1997:** The IBM's **Deep blue** intelligent computer won the chess game against the chess expert Garry Kasparov, and it became the first computer which had beaten a human chess expert.

Machine Learning at present:

The field of machine learning has made significant strides in recent years, and its applications are numerous, including self-driving cars, Amazon Alexa, Catboats, and the recommender system. It incorporates clustering, classification, decision tree, SVM algorithms, and reinforcement learning, as well as unsupervised and supervised learning.

Present day AI models can be utilized for making different expectations, including climate expectation, sickness forecast, financial exchange examination, and so on.

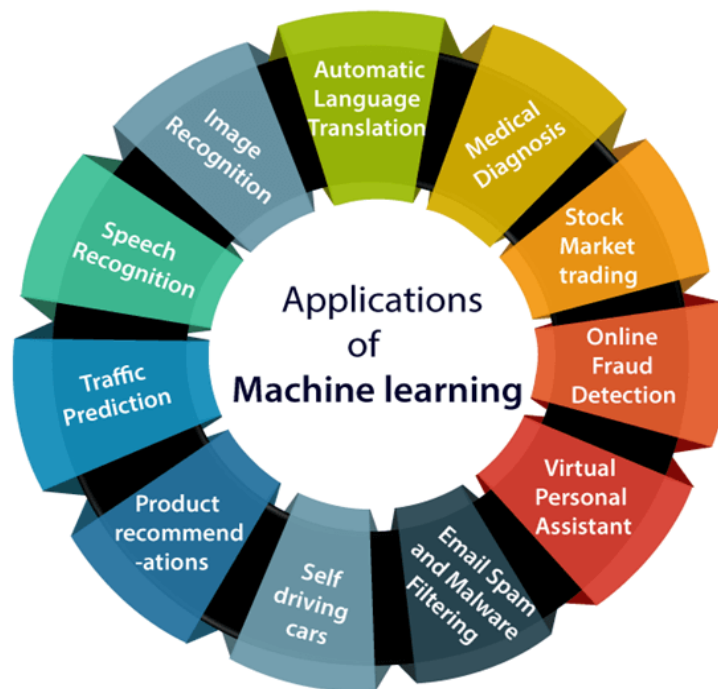
Prerequisites

Before learning machine learning, you must have the basic knowledge of followings so that you can easily understand the concepts of machine learning:

- Fundamental knowledge of probability and linear algebra.
- The ability to code in any computer language, especially in Python language.
- Knowledge of Calculus, especially derivatives of single variable and multivariate functions.

Applications of Machine learning

Machine learning is a buzzword for today's technology, and it is growing very rapidly day by day. We are using machine learning in our daily life even without knowing it such as Google Maps, Google assistant, Alexa, etc. Below are some most trending real-world applications of Machine Learning:



1. Image Recognition:

Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**:

Facebook provides us a feature of auto friend tagging suggestion. Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's **face detection** and **recognition algorithm**.

It is based on the Facebook project named "**Deep Face**," which is responsible for face recognition and person identification in the picture.

2. Speech Recognition

While using Google, we get an option of "**Search by voice**," it comes under speech recognition, and it's a popular application of machine learning.

Speech recognition is a process of converting voice instructions into text, and it is also known as "**Speech to text**", or "**Computer speech recognition**." At

present, machine learning algorithms are widely used by various applications of speech recognition. **Google assistant, Siri, Cortana,** and **Alexa** are using speech recognition technology to follow the voice instructions.

3. Traffic prediction:

If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions.

It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

- **Real Time location** of the vehicle from Google Map app and sensors
- **Average time has taken** on past days at the same time.

Everyone who is using Google Map is helping this app to make it better. It takes information from the user and sends back to its database to improve the performance.

4. Product recommendations:

Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon, Netflix,** etc., for product recommendation to the user. Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

Google understands the user interest using various machine learning algorithms and suggests the product as per customer interest.

As similar, when we use Netflix, we find some recommendations for entertainment series, movies, etc., and this is also done with the help of machine learning.

5. Self-driving cars:

One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering:

Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning. Below are some spam filters used by Gmail:

- Content Filter
- Header filter
- General blacklists filter
- Rules-based filters
- Permission filters

Some machine learning algorithms such as **Multi-Layer Perceptron**, **Decision tree**, and **Naïve Bayes classifier** are used for email spam filtering and malware detection.

7. Virtual Personal Assistant:

We have various virtual personal assistants such as **Google assistant**, **Alexa**, **Cortana**, **Siri**. As the name suggests, they help us in finding the information using our voice instruction. These assistants can help us in various ways just by our voice instructions such as Play music, call someone, Open an email, Scheduling an appointment, etc.

These virtual assistants use machine learning algorithms as an important part.

These assistant record our voice instructions, send it over the server on a cloud, and decode it using ML algorithms and act accordingly.

8. Online Fraud Detection:

Machine learning is making our online transaction safe and secure by detecting fraud transaction. Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction. So to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

For each genuine transaction, the output is converted into some hash values, and these values become the input for the next round. For each genuine transaction, there is a specific pattern which gets change for the fraud transaction hence, it detects it and makes our online transactions more secure.

9. Stock Market trading:

Machine learning is widely used in stock market trading. In the stock market, there is always a risk of up and downs in shares, so for this machine learning's **long short term memory neural network** is used for the prediction of stock market trends.

10. Medical Diagnosis:

In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain.

It helps in finding brain tumors and other brain-related diseases easily.

11. Automatic Language Translation:

Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

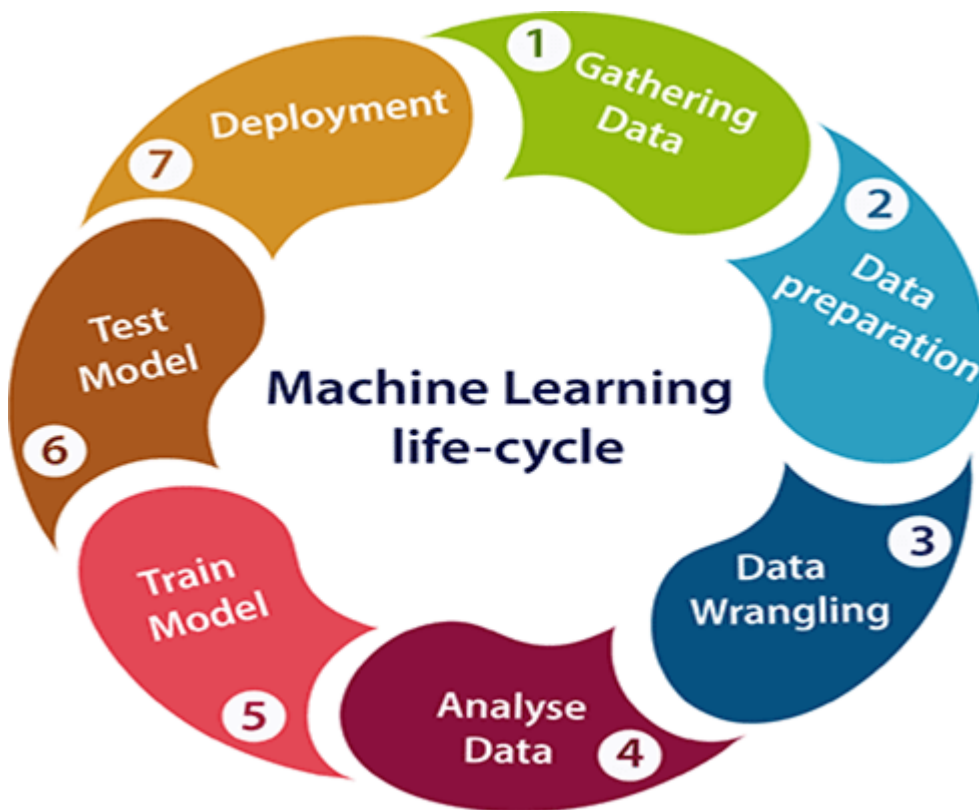
Machine learning Life cycle

Machine learning has given the computer systems the abilities to automatically learn without being explicitly programmed. But how does a machine learning system work? So, it can be described using the life cycle of machine learning. Machine learning life cycle is a cyclic process to build an efficient machine learning project. The main purpose of the life cycle is to find a solution to the problem or project.

Machine learning life cycle involves seven major steps, which are given below:

- **Gathering Data**
- **Data preparation**
- **Data Wrangling**
- **Analyse Data**
- **Train the model**
- **Test the model**

- **Deployment**



The most important thing in the complete process is to understand the problem and to know the purpose of the problem. Therefore, before starting the life cycle, we need to understand the problem because the good result depends on the better understanding of the problem.

In the complete life cycle process, to solve a problem, we create a machine learning system called "model", and this model is created by providing "training". But to train a model, we need data, hence, life cycle starts by collecting data.

1. Gathering Data:

Data Gathering is the first step of the machine learning life cycle. The goal of this step is to identify and obtain all data-related problems.

In this step, we need to identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**. It is one of the most important steps of the life cycle. The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

This step includes the below tasks:

- **Identify various data sources**
- **Collect data**
- **Integrate the data obtained from different sources**

By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

2. Data preparation

After collecting the data, we need to prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.

In this step, first, we put all data together, and then randomize the ordering of data.

This step can be further divided into two processes:

- **Dataexploration:**
It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data. A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.
 - **Datapre-processing**
Now the next step is preprocessing of data for its analysis.
-

3. Data Wrangling

Data wrangling is the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step. It is one of the most important steps of the complete process. Cleaning of data is required to address the quality issues.

It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:

- **Missing Values**
- **Duplicate data**
- **Invalid data**
- **Noise**

So, we use various filtering techniques to clean the data.

It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

4. Data Analysis

Now the cleaned and prepared data is passed on to the analysis step. This step involves:

- **Selection of analytical techniques**
- **Building models**
- **Review the result**

The aim of this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome. It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, etc. then build the model using prepared data, and evaluate the model.

Hence, in this step, we take the data and use machine learning algorithms to build the model.

5. Train Model

Now the next step is to train the model, in this step we train our model to improve its performance for better outcome of the problem.

We use datasets to train the model using various machine learning algorithms. Training a model is required so that it can understand the various patterns, rules, and, features.

6. Test Model

Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.

Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

7. Deployment

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

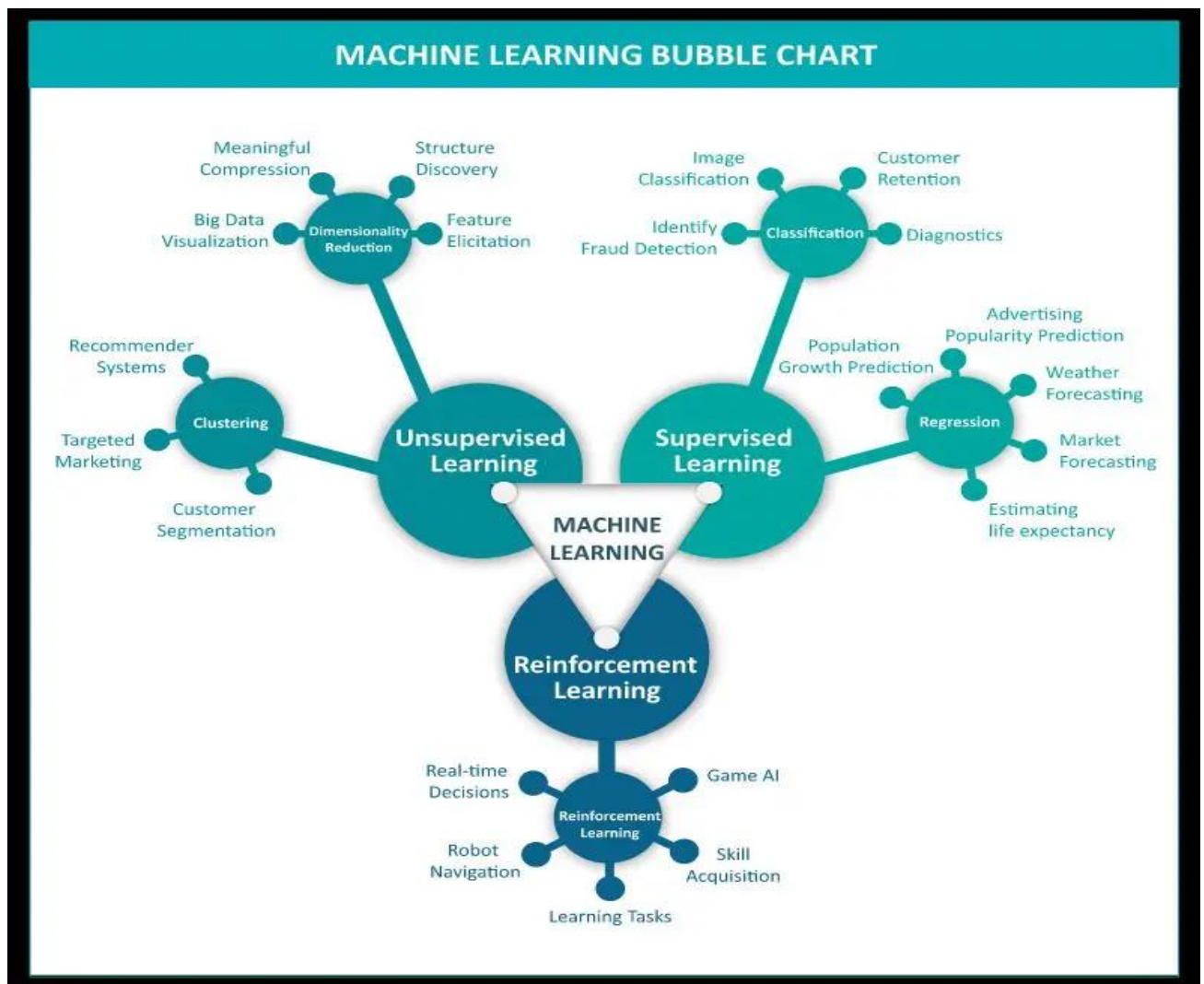
If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

2. Learning paradigms

Machine learning paradigms refer to the different approaches or methodologies used in the field of machine learning to build predictive models and make sense of data. These paradigms encompass various techniques, algorithms, and philosophies for training models and extracting insights from data.

Machine Learning is generally categorized into three primary paradigms. They are:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning



Machine Learning Bubble Chart

1. Supervised Learning:

- **Definition:** In supervised learning, the computer learns from a set of input-output pairs (labeled examples). The goal is to train a predictive model that can guess the output value (label) for new, unseen inputs.
- **Example:** Imagine we have a dataset of animal characteristics. Our goal is to predict an animal's weight based on its other features (such as age and sex). We create input-output pairs, and the computer learns to predict using these examples. For instance:
 - Given input features (age, sex), we train a model to predict the weight of an animal.
 - We can also use supervised learning for text classification (e.g., distinguishing between "cat" and "dog") or image recognition tasks.

- **Phases:** Supervised learning involves two phases:
 - **Training Phase:** The model is trained using labeled examples.
 - **Prediction Phase:** The trained model is used to make predictions for new inputs.

2. Unsupervised Learning:

- **Definition:** In unsupervised learning, there are no labeled outputs. The computer identifies patterns or structures in the data without explicit guidance.
- **Example:** Clustering similar data points together (e.g., grouping customers based on purchasing behavior) or dimensionality reduction (e.g., reducing features while preserving information).
- **Tasks:** Unsupervised learning includes tasks like clustering, anomaly detection, and dimensionality reduction.

3. Reinforcement Learning:

- **Definition:** Reinforcement learning involves an agent interacting with an environment. The agent learns by receiving feedback (rewards or penalties) based on its actions.
- **Example:** Training an AI to play games (e.g., chess, Go) or control robotic systems. The agent explores actions, receives rewards, and adjusts its strategy.
- **Phases:** Reinforcement learning consists of:
 - **Exploration:** The agent explores different actions.
 - **Exploitation:** The agent exploits actions that yield higher rewards.
 - **Learning:** The agent updates its strategy based on rewards.

3. Statistical Decision Theory – Regression and Classification

Statistical decision theory is a framework used to make decisions based on data and probability theory. It provides a systematic approach for making decisions in situations where uncertainty is present. The key components of statistical decision theory include:

1. **Decision Spaces:** Decision theory starts by defining the set of possible actions or decisions that can be taken. These decisions are usually

represented as a set of alternatives or choices available to the decision-maker.

2. **Probability Distributions:** In statistical decision theory, uncertainty is quantified using probability distributions. These distributions represent the likelihood of different outcomes occurring given the available information. Decision theory often involves making decisions under uncertainty, where the outcomes are probabilistic rather than deterministic.
3. **Utility Functions:** Utility functions are used to quantify the desirability or value associated with different outcomes. The utility function assigns a numerical value to each possible outcome, reflecting the decision-maker's preferences or objectives.
4. **Decision Rules:** Decision rules specify how decisions should be made based on the available information. These rules typically involve choosing the decision that maximizes expected utility, which is calculated by combining the probabilities of different outcomes with their associated utilities.
5. **Risk and Uncertainty Management:** Statistical decision theory provides tools for managing risk and uncertainty in decision-making. This includes techniques for assessing the potential consequences of different decisions, as well as strategies for mitigating risks and maximizing rewards.

Statistical decision theory provides a formal framework for making decisions in situations where there is uncertainty or incomplete information. By combining probability theory with decision-making principles, it helps decision-makers make informed choices that are optimized to achieve their objectives.

It involves two main components: classification and regression.

1. **Classification:**

- Classification is a supervised learning task where the goal is to assign a label or category to each input sample.
- In statistical decision theory, classification involves constructing a decision rule or a classifier that maps input features to predefined classes or categories.
- Common classification algorithms include logistic regression, decision trees, support vector machines, and neural networks.

- Evaluation of classification models often involves metrics such as accuracy, precision, recall, F1-score, and ROC curves.

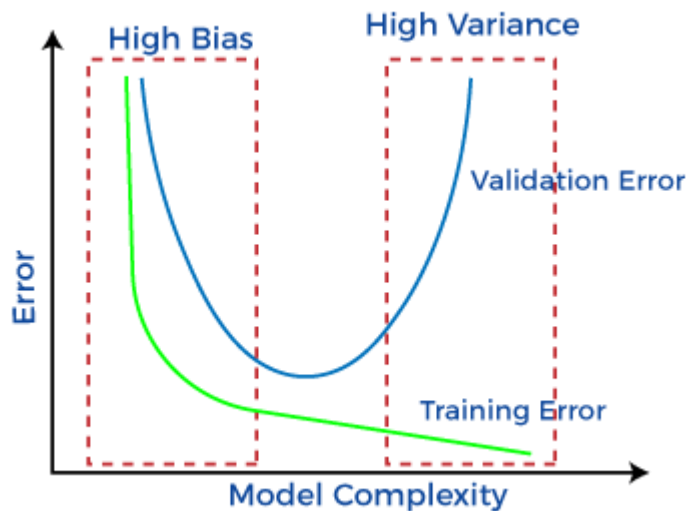
2. Regression:

- Regression is also a supervised learning task, but instead of predicting classes, it predicts continuous numerical values.
- In statistical decision theory, regression involves modeling the relationship between input variables and a continuous target variable.
- The goal is to find a function that maps input features to the target variable, minimizing the difference between predicted and actual values.
- Regression algorithms include linear regression, polynomial regression, decision trees, support vector regression, and neural networks.
- Evaluation of regression models typically involves metrics such as mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R-squared.

Both classification and regression play crucial roles in solving various real-world problems in fields such as finance, healthcare, marketing, and more. They form the backbone of many machine learning applications, helping to extract insights and make predictions from data.

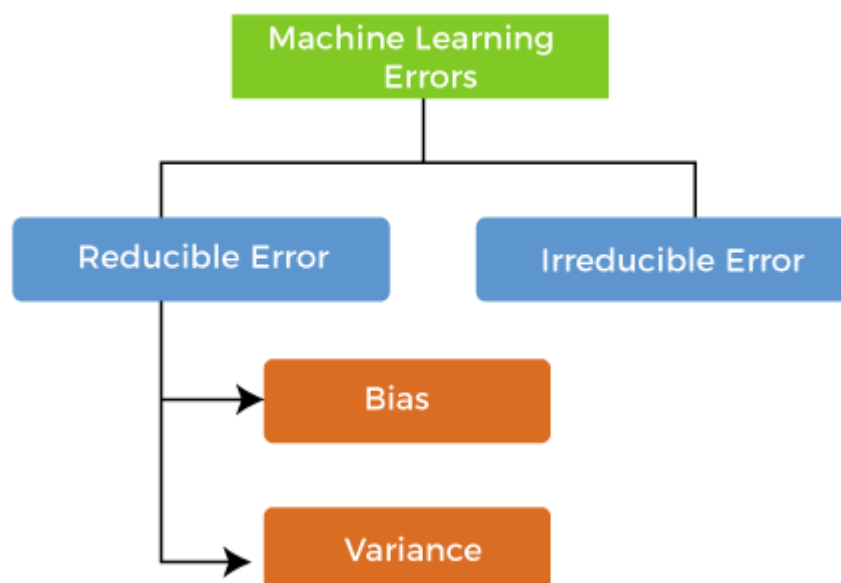
4 .Bias-Variance

Machine learning is a branch of Artificial Intelligence, which allows machines to perform data analysis and make predictions. However, if the machine learning model is not accurate, it can make prediction errors, and these prediction errors are usually known as Bias and Variance. In machine learning, these errors will always be present as there is always a slight difference between the model predictions and actual predictions. The main aim of ML/data science analysts is to reduce these errors in order to get more accurate results.



Errors in Machine Learning?

In machine learning, an error is a measure of how accurately an algorithm can make predictions for the previously unknown dataset. On the basis of these errors, the machine learning model is selected that can perform best on the particular dataset. There are mainly two types of errors in machine learning, which are:



- **Reducible errors:** These errors can be reduced to improve the model accuracy. Such errors can further be classified into bias and Variance.
- **Irreducible errors:** These errors will always be present in the model regardless of which algorithm has been used. The cause of these errors is unknown variables whose value can't be reduced.

What is Bias?

In general, a machine learning model analyses the data, find patterns in it and make predictions. While training, the model learns these patterns in the dataset and applies them to test data for prediction. ***While making predictions, a difference occurs between prediction values made by the model and actual values/expected values, and this difference is known as bias errors or Errors due to bias.*** It can be defined as an inability of machine learning algorithms such as Linear Regression to capture the true relationship between the data points. Each algorithm begins with some amount of bias because bias occurs from assumptions in the model, which makes the target function simple to learn. A model has either:

- **Low Bias:** A low bias model will make fewer assumptions about the form of the target function.
- **High Bias:** A model with a high bias makes more assumptions, and the model becomes unable to capture the important features of our dataset. **A high bias model also cannot perform well on new data.**

Ways to reduce High Bias:

High bias mainly occurs due to a much simple model. Below are some ways to reduce the high bias:

- Increase the input features as the model is underfitted.
- Decrease the regularization term.
- Use more complex models, such as including some polynomial features.

What is a Variance Error?

The variance would specify the amount of variation in the prediction if the different training data was used. In simple words, ***variance tells that how much a random variable is different from its expected value.*** Ideally, a model should not vary too much from one training dataset to another, which means the algorithm should be good in understanding the hidden mapping between inputs and output variables. Variance errors are either of **low variance or high variance.**

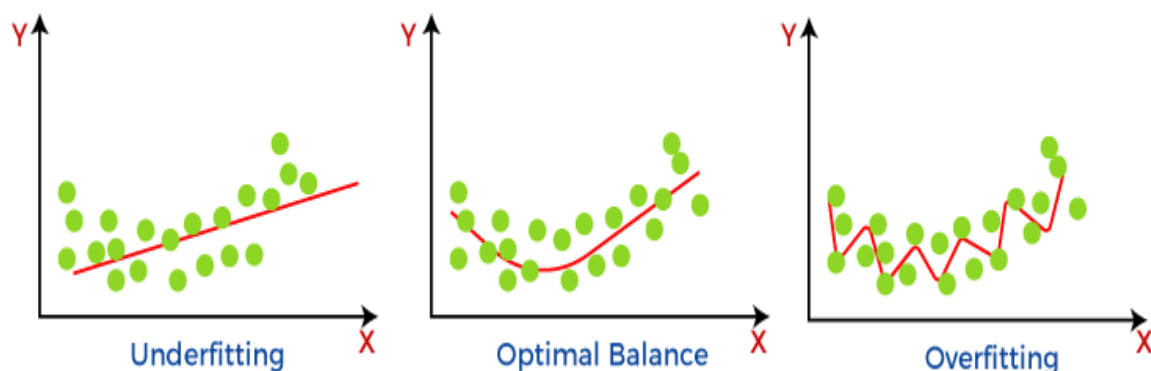
Low variance means there is a small variation in the prediction of the target function with changes in the training data set. At the same time, **High variance** shows a large variation in the prediction of the target function with changes in the training dataset.

A model that shows high variance learns a lot and perform well with the training dataset, and does not generalize well with the unseen dataset. As a result, such a model gives good results with the training dataset but shows high error rates on the test dataset.

Since, with high variance, the model learns too much from the dataset, it leads to overfitting of the model. A model with high variance has the below problems:

- A high variance model leads to overfitting.
- Increase model complexities.

Usually, nonlinear algorithms have a lot of flexibility to fit the model, have high variance.



Some examples of machine learning algorithms with low variance are, **Linear Regression, Logistic Regression, and Linear discriminant analysis**. At the same time, algorithms with high variance are **decision tree, Support Vector Machine, and K-nearest neighbours**.

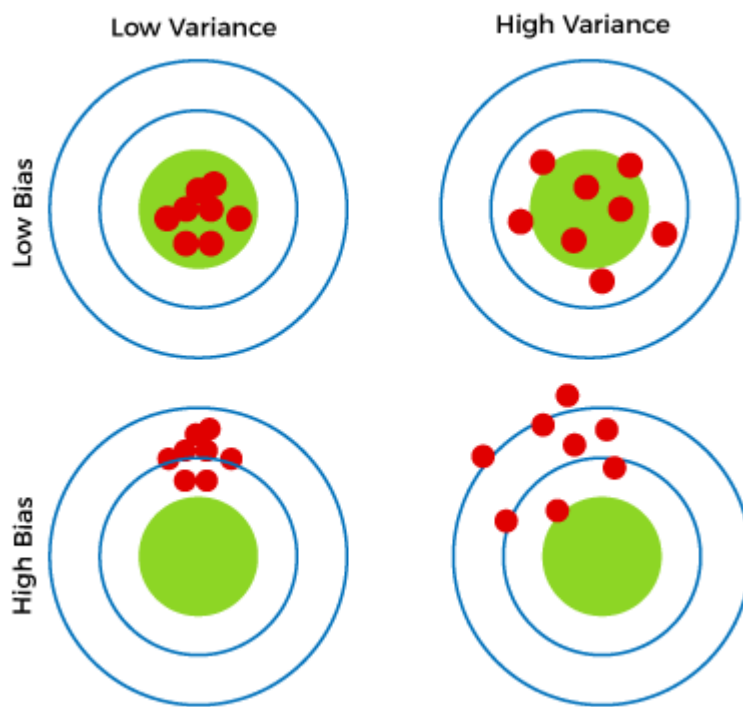
Ways to Reduce High Variance:

- Reduce the input features or number of parameters as a model is overfitted.
- Do not use a much complex model.
- Increase the training data.

- Increase the Regularization term.

Different Combinations of Bias-Variance

There are four possible combinations of bias and variances, which are represented by the below diagram:



1. **Low-Bias,Low-Variance:**

The combination of low bias and low variance shows an ideal machine learning model. However, it is not possible practically.

2. **Low-Bias, High-Variance:** With low bias and high variance, model predictions are inconsistent and accurate on average. This case occurs when the model learns with a large number of parameters and hence leads to an **overfitting**

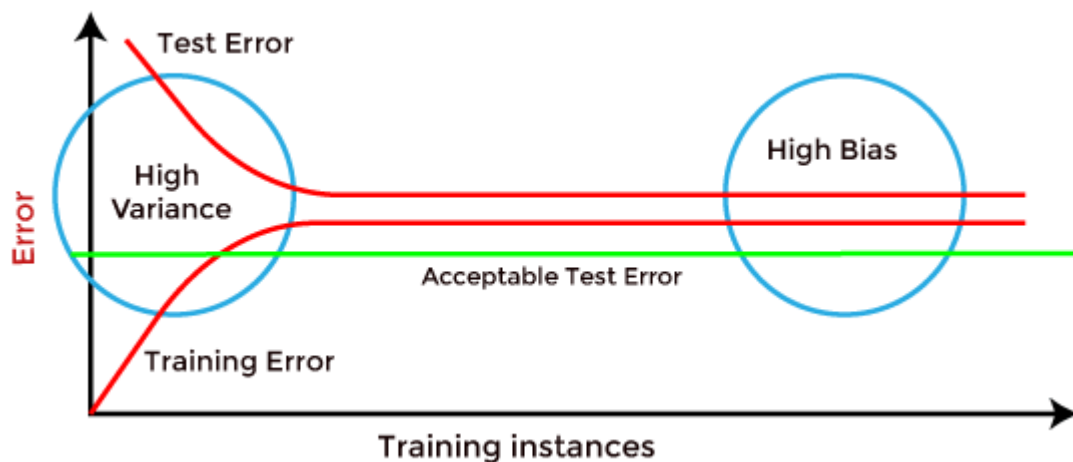
3. **High-Bias, Low-Variance:** With High bias and low variance, predictions are consistent but inaccurate on average. This case occurs when a model does not learn well with the training dataset or uses few numbers of the parameter. It leads to **underfitting** problems in the model.

4. **High-Bias,High-Variance:**

With high bias and high variance, predictions are inconsistent and also inaccurate on average.

How to identify High variance or High Bias?

High variance can be identified if the model has:



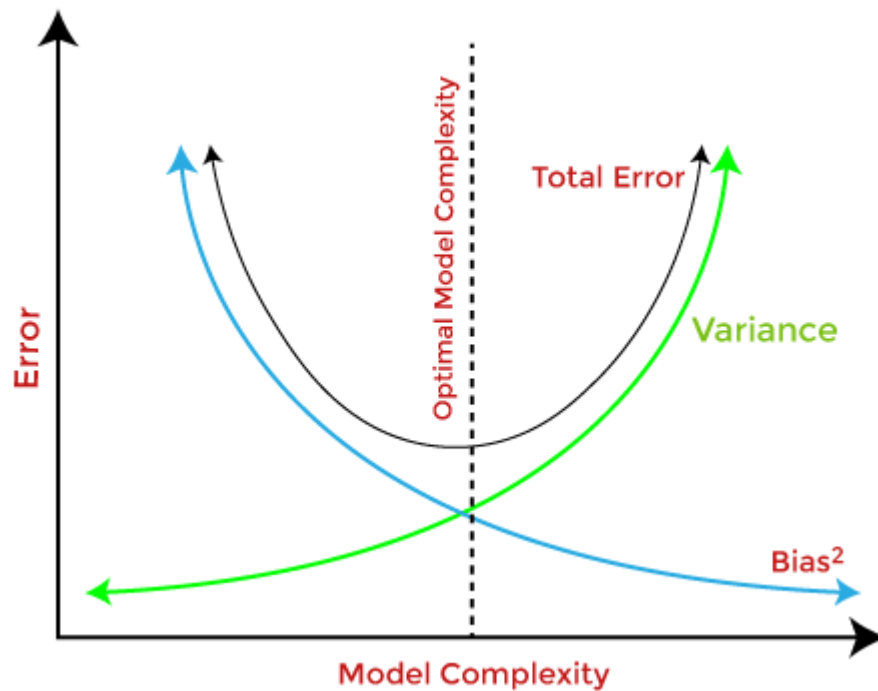
- Low training error and high test error.

High Bias can be identified if the model has:

- High training error and the test error is almost similar to training error.

Bias-Variance Trade-Off

While building the machine learning model, it is really important to take care of bias and variance in order to avoid overfitting and underfitting in the model. If the model is very simple with fewer parameters, it may have low variance and high bias. Whereas, if the model has a large number of parameters, it will have high variance and low bias. So, it is required to make a balance between bias and variance errors, and this balance between the bias error and variance error is known as **the Bias-Variance trade-off**.



For an accurate prediction of the model, algorithms need a low variance and low bias. But this is not possible because bias and variance are related to each other:

- If we decrease the variance, it will increase the bias.
- If we decrease the bias, it will increase the variance.

Bias-Variance trade-off is a central issue in supervised learning. Ideally, we need a model that accurately captures the regularities in training data and simultaneously generalizes well with the unseen dataset. Unfortunately, doing this is not possible simultaneously. Because a high variance algorithm may perform well with training data, but it may lead to overfitting to noisy data. Whereas, high bias algorithm generates a much simple model that may not even capture important regularities in the data. So, we need to find a sweet spot between bias and variance to make an optimal model.